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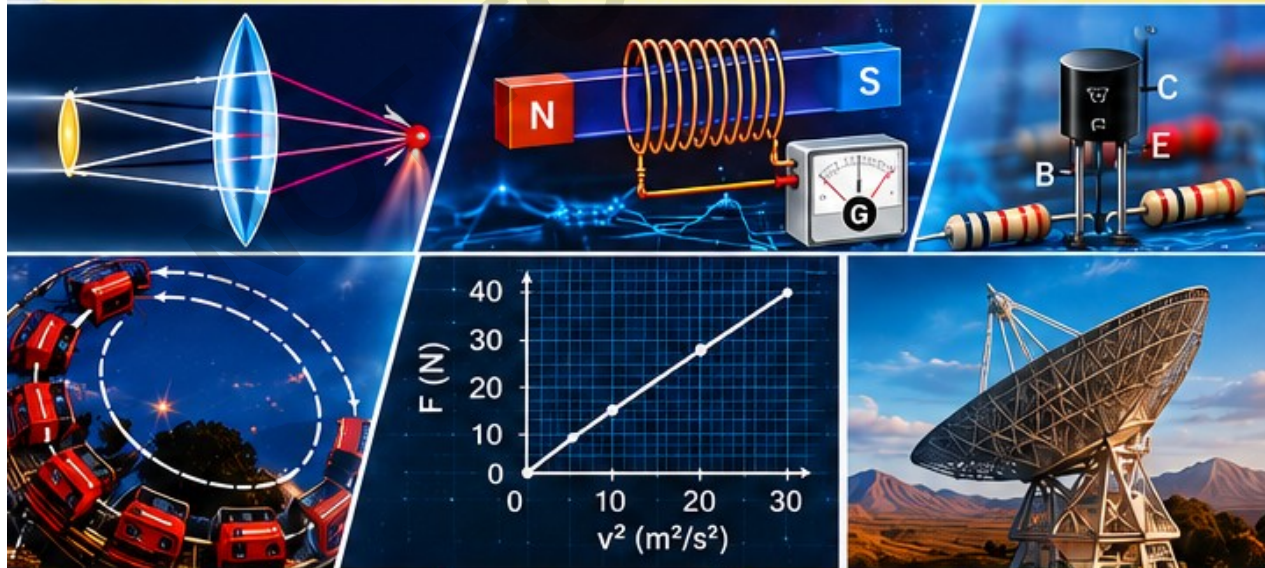
GRADE 12

PHYSICS

REVISION BOOK 2

Practice • Application • Mastery

Topical Practice | Mixed Concepts | Practical & Data Analysis
6 CBE National Examination Test Models | Complete Marking Guidance



**TOPICAL
PRACTICE**



**MIXED
CONCEPTS**



**PRACTICAL &
DATA ANALYSIS**



**6 CBE TEST
MODELS**



**COMPLETE
MARKING
GUIDANCE**

HEZEKIAH OMWERI NYATUKA

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PREFACE

Grade 12 Physics Revision Book 2 is the practice and mastery companion to Grade 12 Physics Revision Book 1. Book 2 places greater emphasis on practice, application, investigation, interpretation and problem solving.

REVISE → ATTEMPT → APPLY → CHECK → CORRECT → MASTER

How to Use This Book

- **REVISE** — begin with the short revision points and equations.
- **ATTEMPT** — try every question before consulting the answers.
- **APPLY** — connect Physics concepts to practical and real-life situations.
- **CHECK** — compare your method and answer with the marking guidance.
- **CORRECT** — identify the reason for every mistake and correct it.
- **MASTER** — repeat difficult questions until you can solve them independently.

REMEMBER: Understanding why an answer is correct is more valuable than memorising the answer.
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CONTENTS

- **PART I — TOPICAL REVISION & MASTERY:** Dynamics; Uniform Circular Motion; Floating and Sinking; Quantity of Heat; Thin Lenses; Photoelectric Effect; Electromagnetic Induction; Mains Electricity; Transistors; Introduction to Satellites.
- **PART II — MIXED CONCEPTS:** Sets 1–3 integrating Grades 9, 10, 11 and 12 Physics.
- **PART III — PRACTICAL, INVESTIGATION, GRAPH & DATA-ANALYSIS MASTERY:** Sets 1–2.
- **PART IV — CBE NATIONAL EXAMINATION TEST MODELS:** Models 1–6, 50 marks each, with answers and marking guidance.
- **PART V — FINAL QUICK-REFERENCE SECTION:** formulas, constants, units, graph guide, common errors, checklist and exam tips.

PART I — TOPICAL REVISION & MASTERY

1.1 Dynamics

Quick Revision

$$p = mv$$

$$J = F\Delta t = \Delta p$$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Topical Mastery

1. A 0.50 kg ball changes velocity from +12 m/s to -8 m/s in 0.040 s. Find the change in momentum and average force.
2. Two trolleys collide and stick together. Use conservation of momentum to determine their common velocity.
3. Explain how airbags and crumple zones reduce injury in terms of impulse.

EXAM FOCUS: Use Formula → Substitution → Working → Final Answer + Unit. For vector quantities, include direction.

COMMON ERROR: Keep units consistent and distinguish scalar quantities from vector quantities where relevant.

1.2 Uniform Circular Motion

Quick Revision

$$\omega = 2\pi f = 2\pi/T$$

$$v = r\omega$$

$$a_c = v^2/r$$

$$F_c = mv^2/r$$

Topical Mastery

1. A 2.0 kg mass moves at 6.0 m/s in a circle of radius 3.0 m. Find centripetal acceleration and force.
2. Investigate how centripetal force varies with speed using a rotating stopper.
3. Explain why a vehicle requires an inward resultant force when negotiating a bend.

EXAM FOCUS: Use Formula → Substitution → Working → Final Answer + Unit. For vector quantities, include direction.

COMMON ERROR: Keep units consistent and distinguish scalar quantities from vector quantities where relevant.

1.3 Floating and Sinking

Quick Revision

$$\rho = m/V$$

$$U = \rho_{\text{fluid}} V_{\text{displaced}} g$$

$$\text{For floating equilibrium: } U = W$$

PART II — MIXED CONCEPTS (GRADES 9–12)

Mixed Concepts Set 1

Attempt each task using relevant Physics from earlier grades and Grade 12. Show calculations and justify explanations.

1. Road safety: relate momentum, impulse and stopping time to passenger protection.
2. Raft: apply density, upthrust and flotation to determine safe loading.
3. Heater: calculate electrical energy and resulting temperature change.
4. Projector: use the thin-lens equation and magnification.
5. Motorcycle on a circular track: calculate centripetal force.
6. Bicycle generator: explain electromagnetic induction.
7. Home mains: calculate current/energy and explain safety devices.
8. Greenhouse fan: explain transistor switching using a temperature sensor.
9. Photoelectric sensor: apply threshold frequency and stopping potential.
10. Earth-observation satellite: calculate orbital speed and state an environmental use.
11. Heating graph: interpret a plateau and latent heat.
12. Solar-powered monitoring station: integrate electrical power, sensors and data collection.

THINK BIG: Where a problem combines several topics, identify the principle required at each stage before calculating.
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Mixed Concepts Set 2

Attempt each task using relevant Physics from earlier grades and Grade 12. Show calculations and justify explanations.

1. Delivery van: momentum and braking.
2. Floating research instrument: density and fraction submerged.
3. Solar heating: energy transfer and specific heat capacity.
4. Camera: image formation by a convex lens.
5. Road bend: circular motion and safe speed.
6. Laboratory heater: electrical power and heating.
7. Wind generator: electromagnetic induction.
8. Specific-heat-capacity investigation: variables, data and errors.
9. Heating/phase-change graph: identify regions and energy changes.
10. Transistor alarm: sensor response and switching.
11. Photoelectric light detector: photon energy and threshold behaviour.

PART III — PRACTICAL, INVESTIGATION, GRAPH & DATA-ANALYSIS MASTERY

Practical Mastery Set 1

1. Momentum trolley collision: compare total momentum before and after collision.
2. Circular motion with a rubber stopper: investigate F against v^2 .
3. Archimedes investigation: compare apparent loss of weight with upthrust.
4. Specific heat capacity of an electrical block: determine c from $VIt = mc\Delta T$.
5. Thin lens focal length: obtain repeated u and v readings.
6. Photoelectric graph: analyse $K_{\max}$ against f .
7. Electromagnetic induction: compare magnet speed with induced e.m.f.
8. Low-voltage V – I investigation: tabulate and interpret current/voltage data.
9. LDR data: investigate resistance against light intensity.
10. Satellite data: test the relationship T^2/r^3 .
11. Graph skills: scales, best-fit lines, gradients and intercepts.
12. Experimental errors: distinguish random and systematic effects.
13. Peak challenge: interpret unfamiliar V – I data and evaluate method.

PRACTICAL CHECK: For every investigation identify: aim, apparatus, independent variable, dependent variable, controlled variables, procedure, data table, graph/analysis, conclusion, precaution and improvement.

Practical Mastery Set 2

1. Complete missing values in an electrical-power data table.
2. Interpret a velocity–time graph and determine motion quantities.
3. Identify anomalous data in a heating experiment.
4. Plot $1/v$ against $1/u$ for a thin lens and interpret the graph.
5. Determine the density of an unknown liquid using upthrust.
6. Transform circular-motion data to test $F \propto v^2$.
7. Design an induction investigation with clear variables.
8. Analyse transformer input/output data and efficiency.
9. Analyse NTC thermistor resistance–temperature data.
10. Use a photoelectric graph to estimate threshold frequency.
11. Use a T^2 against r^3 satellite graph to test orbital theory.

PART IV — CBE NATIONAL EXAMINATION TEST MODELS

CBE NATIONAL EXAMINATION TEST MODEL 1

Time: 1 Hour 30 Minutes | Total: 50 Marks

This is an independently developed CBE national-examination-style practice test and is not an official national examination paper.

SECTION A — SHORT RESPONSE (15 marks)

1. Define momentum and state its SI unit. (2)
2. State the direction of centripetal acceleration. (1)
3. State the condition for a body floating at rest. (1)
4. Distinguish specific heat capacity from specific latent heat. (2)
5. State the image characteristics for an object between F and $2F$ of a convex lens. (2)
6. Define threshold frequency. (1)
7. State Faraday's law of electromagnetic induction. (2)
8. Give one reason household appliances are connected in parallel. (1)
9. State the relation between transistor current gain, collector current and base current. (1)
10. State two conditions for a geostationary satellite. (2)

SECTION B — STRUCTURED & CALCULATION (20 marks)

11. A 1200 kg car moving at 20 m/s is brought to rest in 4.0 s. Calculate initial momentum, change in momentum and average braking force; explain one road-safety application. (8)
12. A 1.5 kW heater warms 2.0 kg of water for 5.0 min. Assuming no losses and $c = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$, calculate energy supplied and temperature rise; state one reason the actual rise may be smaller. (6)
13. A 400-turn coil experiences a flux change from $6.0 \times 10^{-4} \text{ Wb}$ to $2.0 \times 10^{-4} \text{ Wb}$ in 0.080 s. Calculate induced e.m.f.; state how to increase it, explain Lenz's law, and state when no e.m.f. is induced. (6)

SECTION C — APPLICATION, PRACTICAL & DATA ANALYSIS (15 marks)

14. Explain how an NTC thermistor, potential divider and transistor can control a greenhouse fan. (5)
15. A metal has $f_0 = 4.0 \times 10^{14} \text{ Hz}$ and is illuminated by $f = 7.0 \times 10^{14} \text{ Hz}$. Calculate work function, maximum kinetic energy and stopping potential. Use $h = 6.63 \times 10^{-34} \text{ J s}$, $e = 1.60 \times 10^{-19} \text{ C}$. (5)
16. A satellite orbits Earth at $r = 8.0 \times 10^6 \text{ m}$. Calculate orbital speed using $G = 6.67 \times 10^{-11}$ and $M = 6.0 \times 10^{24} \text{ kg}$; state what happens to period in a higher orbit and give one application. (5)

CBE NATIONAL EXAMINATION TEST MODEL 1 — ANSWERS & MARKING GUIDANCE

11. $p = 2.40 \times 10^4 \text{ kg m/s}$; $\Delta p = -2.40 \times 10^4 \text{ kg m/s}$; $F = -6.0 \times 10^3 \text{ N}$. Increasing stopping time reduces force.
12. $E = Pt = 4.50 \times 10^5 \text{ J}$; $\Delta T = E/(mc) \approx 53.6 \text{ K}$. Actual rise is lower because of heat losses.